SEQUENCES LADDER 6 OF 9

Falling factorials: the powers of finite calculus

The products $n(n-1)(n-2)\cdots$ difference as neatly as x^k differentiates.

1 / THE RULE AND THE PICTURE

The difference of $n^{\underline{k}}$ is $k\,n^{\underline{k-1}}$

$n^{\underline{3}}$	0	0	0	6	24	60	120
Δ		0	0	6	18	36	60

$\Delta n^{\underline{3}} = 3n^{\underline{2}}$: $0, 0, 6, 18, 36, 60$

Write $n^{\underline{k}} = n(n-1)\cdots(n-k+1)$, a falling factorial with k factors. Then $(n+1)^{\underline{k}} - n^{\underline{k}} = n^{\underline{k-1}}\big[(n+1)-(n-k+1)\big]$, so

$$\boxed{\Delta\, n^{\underline{k}} = k\, n^{\underline{k-1}}}$$

the exact copy of $\frac{d}{dx}x^k = kx^{k-1}$, with no extra terms.

2 / THE TABLE OF VALUES

Each row is the difference of the row below, times the index

$n =$	0	1	2	3	4	5	6
$n^{\underline{1}}$	0	1	2	3	4	5	6
$n^{\underline{2}}$	0	0	2	6	12	20	30
$n^{\underline{3}}$	0	0	0	6	24	60	120
$n^{\underline{4}}$	0	0	0	0	24	120	360

For example the row of $n^{\underline{3}}$ has differences $0, 0, 6, 18, 36, 60$, which is 3 times the row of $n^{\underline{2}}$, namely $3\cdot 0, 3\cdot 0, 3\cdot 2, 3\cdot 6, 3\cdot 12, 3\cdot 20$.

3 / CHANGING BETWEEN POWERS AND FALLING FACTORIALS

The coefficients are the Stirling numbers of the second kind

Every power is a sum of falling factorials with whole-number coefficients: $n^2 = n^{\underline{2}} + n^{\underline{1}}$, $n^3 = n^{\underline{3}} + 3n^{\underline{2}} + n^{\underline{1}}$ and $n^4 = n^{\underline{4}} + 6n^{\underline{3}} + 7n^{\underline{2}} + n^{\underline{1}}$. A check at $n = 4$ gives $256 = 24 + 144 + 84 + 4$, and the numbers $1, 6, 7, 1$ are a row of the Stirling numbers of the second kind.

4 / SUMMING IS AS EASY AS INTEGRATING

$$\sum_{i=0}^{n-1} i^{\underline{k}} = \frac{n^{\underline{k+1}}}{k+1}$$

Because $\Delta\frac{n^{\underline{k+1}}}{k+1} = n^{\underline{k}}$, adding the terms up to $n-1$ collapses to $\frac{n^{\underline{k+1}}}{k+1}$, just as $\int x^k\,dx = \frac{x^{k+1}}{k+1}$.

To add squares, write $i^2 = i^{\underline{2}} + i^{\underline{1}}$ and sum: $\sum_{i=0}^{n-1} i^2 = \frac{n^{\underline{3}}}{3} + \frac{n^{\underline{2}}}{2}$, which at $n = 5$ is $20 + 10 = 30 = 0 + 1 + 4 + 9 + 16$.

5 / THE DICTIONARY BEGINS

Differences and derivatives share a rule

A falling factorial plays the part of a power, the difference plays the part of the derivative, and the sum $\sum_{i=0}^{n-1}$ plays the part of the integral from 0 to n.

The formulae look identical, which is why finite differences make a gentle first course in calculus.

6 / TRY THESE

Use the rules

(a) Find $\Delta\big(n(n-1)(n-2)\big) = 3n(n-1)$, which at $n = 4$ is 36, and indeed $60 - 24 = 36$ for $n = 4 \to 5$. **(b)** Write n^3 as falling factorials and sum: $\sum_{i=0}^{n-1} i^3 = \frac{n^{\underline{4}}}{4} + n^{\underline{3}} + \frac{n^{\underline{2}}}{2}$, which at $n = 4$ is $6 + 24 + 6 = 36 = 0 + 1 + 8 + 27$. **(c)** Check $\Delta(n^2) = 2n + 1$ with $n^2 = n^{\underline{2}} + n$: $2n + 1$.

ASIDE / STIRLING NUMBERS

The coefficients $1, 3, 1$ and $1, 6, 7, 1$ count the ways of splitting a set into blocks, and multiplying by $k!$ gives the Worpitzky numbers $1, 14, 36, 24$ of the interpolation sheet.

LINKS TO OTHER SHEETS

The previous sheet gives sums of powers by differences; the next sheets let the step between terms shrink to zero.

Convention. $n^{\underline{k}} = n(n-1)\cdots(n-k+1)$ is a falling factorial; $\Delta f(n) = f(n+1) - f(n)$; the sum runs from $i = 0$ to $n-1$.